Sayfa 222 Şekil 4.29

M-PSk

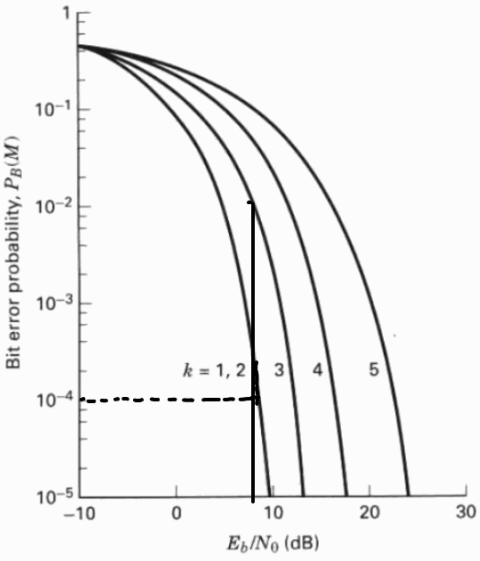


Figure 4.29 Bit error probability for coherently detected multiple phase signaling.

Markkça bit hata olasılığı artıyor

M↑ ⇒ P_D↑ performans kötüleşiyor

x BG aynı (BBV↑) → BG verimli teknik

$BGV = \frac{R_D \text{ bit/sn}}{BG \ H_z}$ ↑

BG verimi

MQAM ✓

MASk ✓

QPSk 2bit — BPSk 1bit dan avantajlıdır

aynı hata olasılığı

Sayfa 230 Şekil 4.35

Gray kodlama ⇒ $P_D \approx \frac{1}{k} P_S$

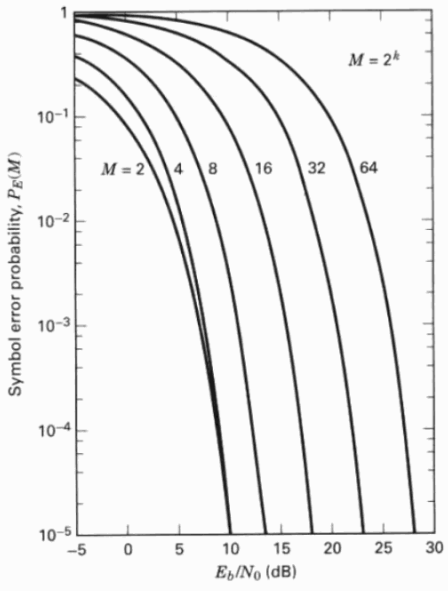
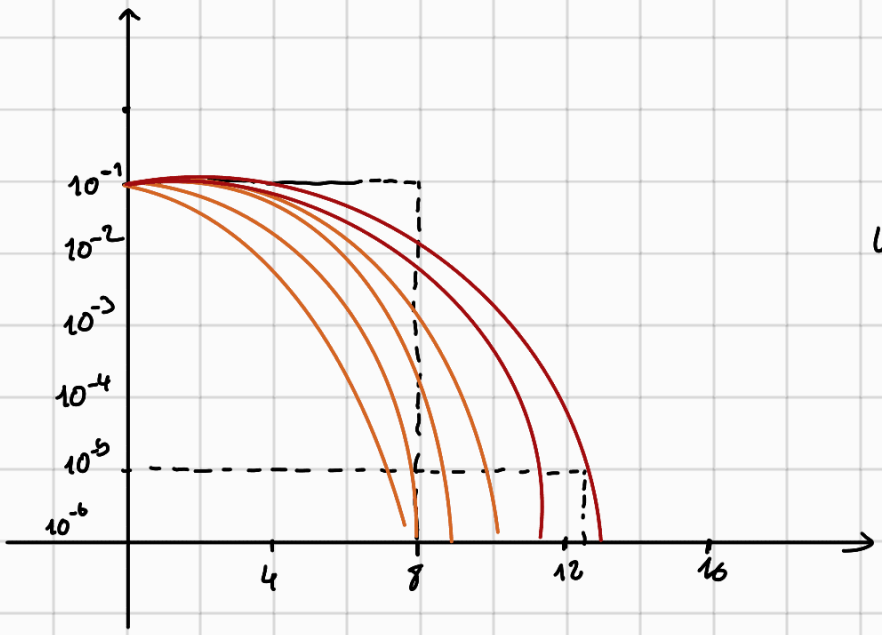


Figure 4.35 Symbol error probability for coherently detected multiple phase signaling. (Reprinted from W. C. Lindsey and M. K. Simon, *Telecommunication Systems Engineering*, Prentice-Hall, Inc. Englewood Cliffs, N.J., 1973, courtesy of W. C. Lindsey and Marvin K. Simon.)

Sayfa 221, Şekil 4.28 (M-FSk)

MFSk

MFSK



$M\uparrow \Rightarrow P_b\downarrow \times BG\uparrow (BGV\downarrow)$

Performans iyileşir — Güç verimi

Uydu haberleşmesinde kullanılır

Ortogonal modülasyon teknikleri için

$$P_b = \frac{M}{2(M-1)} P_s$$

MFSK — Gücün önemli olduğu Bant genişliği önemsenmediği yerde MPSK'ya tercih edilir.

Performans ifadeleri

2-PSK (BPSK), 4-PSK (QPSK) } $P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$

2-FSK (BFS) — Faz tahminli d. / Coherent edemez: $P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$ $\frac{1}{2T}$ → Bant genişliği bakımından daha avantajlı.

Faz tahminsiz (Non coherent): $P_b = \frac{1}{2} e^{-\frac{E_b}{2N_0}}$ $\frac{1}{T}$

$$Q(x) \le e^{-x^2/2}$$

MPSK

For large SNRs and $M \ge 8$

$$P_s \cong 2Q\left(\sqrt{\frac{2E_s}{N_0}} \sin\left(\frac{\pi}{M}\right)\right)$$

, $E_s = k \cdot E_b$ → Bit enerjisi; E_s: sembol enerjisi

→ Gray kodlama → $P_b \approx \frac{1}{k} P_s$

MFSK

Faz tahmin hatasız geçerli

$$P_s \le (M-1)\, Q\left(\sqrt{\frac{E_s}{N_0}}\right)$$

$$P_b = \frac{M}{2(M-1)} P_s \Rightarrow P_b \le \frac{M}{2} Q\left(\sqrt{\frac{E_s}{N_0}}\right) \xrightarrow{k E_b}$$

Faz tahminsiz $\quad P_s \le (M-1)\left(\frac{1}{2} e^{-\frac{E_b}{2N_0}}\right)$

Sayfa 240 Prk 4.16

Yalnızca P_b dikkate alınırsa hangisini tercih edersiniz

Coherent 8 FSK, $\frac{E_b}{N_0} = 8$ dB

// 8-PSK, $\frac{E_b}{N_0} = 13$ dB, gray coded $\rightarrow \frac{E_b}{N_0} = 10^{1,3} = 19,952$

$10 \log_{10} \frac{E_b}{N_0} = 8 \rightarrow \frac{E_b}{N_0} = 10^{0,8} = 6,309$

8FSK: $P_b \le \frac{M}{2} Q\left(\sqrt{k \frac{E_b}{N_0}}\right) = \frac{7}{2} Q\left(\sqrt{3 \cdot 6,309}\right) = 27,2 \cdot 10^{-6} \approx 3 \cdot 10^{-5}$

$M = 8 = 2^3 \Rightarrow k = 3$ bit/sembol

$6,8 \cdot 10^{-6}$

8PSK: $P_b = \frac{1}{k} P_s \rightarrow 3 = \frac{1}{3} 2 Q\left(\sqrt{2k \frac{E_b}{N_0} \sin\left(\frac{\pi}{8}\right)}\right) = 0,886 \cdot 10^{-5} \simeq 10^{-5}$

$\frac{E_b}{N_0} \downarrow 19,952$, $4,187$, $1,33 \cdot 10^{-5}$

8PSK'de hata daha az olduğundan tercih ederiz

Örnek: S.238, Prk 4.7

BFSK, coherent detection, $P(s_1) = P(s_2) = \frac{1}{2}$

$s_1(t) = 0,5 \cos(2000\pi t) \rightarrow f_1 = 1000$ Hz

$s_2(t) = 0,5 \cos(2020\pi t) \rightarrow f_2 = 1010$ Hz

$f_2 - f_1 = 10$ Hz ?? ortogonallik şartı $\Delta f = \frac{1}{2T} = 50$ Hz

$\Rightarrow$ ortogonal değil

AWGN, $\frac{N_0}{2} = 0,001 \Rightarrow P_b = Q\left(\sqrt{\frac{E_b}{N_0}(1-\rho)}\right)$

$T_0 = 0,01$ s

$P_b = ?$

$E_b = E_{s1} P(s_1) + E_{s2} P(s_2)$

$E_{s1} = E_{s2} = (0,5^2) \cdot \frac{1}{2} \cdot T_{(0,01)} = 0,00125 = 1,25$ mJ

$\Rightarrow E_b = 0,00125$ J

$$\rho = \frac{1}{E_b} \int_0^T s_1(t) s_2(t)\, dt = \frac{1}{0,00125} \int_0^{0,01} 0,5 \cos(2000\pi t)\, 0,5 \cos(2020\pi t)\, dt$$

$$= \frac{0,5^2}{0,00125} \frac{1}{2} \int_0^{0,01} [\cos(20\pi t) + \cos(4020\pi t)]\, dt$$

$\rho = 0,94 \rightarrow P_b = Q\left(\sqrt{\frac{0,00125}{2.0,0001}(1-0,94)}\right) = Q(2,743) \approx 0,3 = 3.10^{-1}$

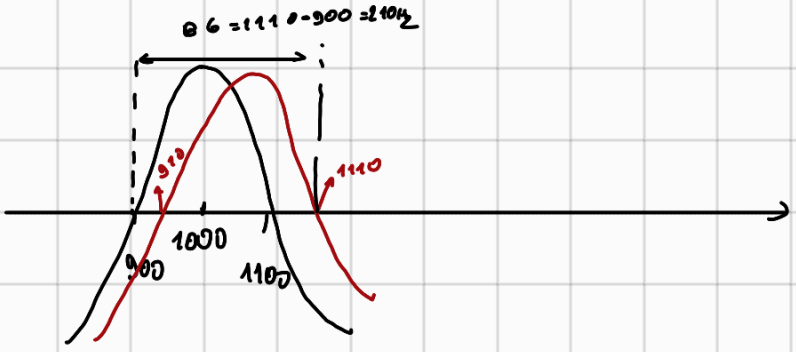
10 bitte 3 bit net o

Ortogonal olsaydı

$P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right) = Q\left(\sqrt{\frac{0,00125}{2.0,0001}}\right) = Q(2,5) = 0,0062 \approx 6.10^{-3}$

Performansı 200 kat iyileştirdik

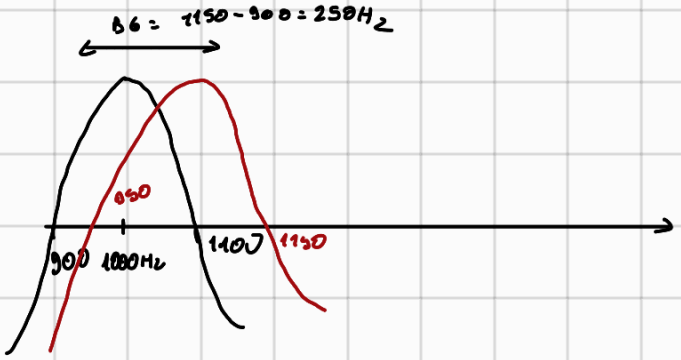
BG Ortogonal değilken $\frac{1}{T} = 100$



Ortogonaliken

$\Delta f = \frac{1}{2T} = 50Hz \rightarrow f_1 = 1000Hz$

$f_2 = 1050\ Hz$



Örnek S.235, Pr. 4,8

4.8. Find the optimum (minimum probability of error) threshold γ_0, for detecting the equally likely signals $s_1(t) = \sqrt{2E/T}\cos\omega_0 t$ and $s_2(t) = \sqrt{\frac{1}{2}E/T}\cos(\omega_0 t + \pi)$ in AWGN, using a correlator receiver as shown in Figure 4.7b. Assume a reference signal of $\psi_1(t) = \sqrt{2/T}\cos\omega_0 t$.

$s_1(t) = \sqrt{\frac{2E}{T}}\cos(\omega_0 t)$

$s_2(t) = \sqrt{\frac{2E}{T}}\cos(\omega_0 t + \pi)$

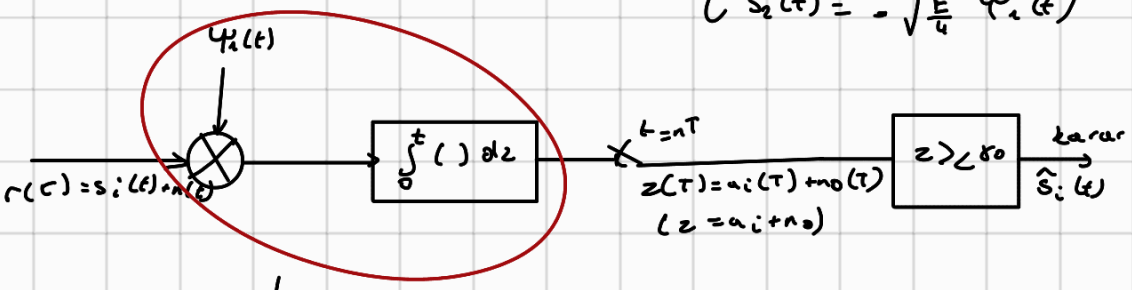
$P(s_1) = P(s_2)$

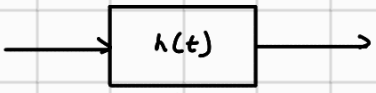
2 QAM gibi

Ref signal $Y_1(t) = \sqrt{\frac{2}{T}}\cos(\omega_0 t)$

$\gamma_0 = ?$

$$\psi_1(t) = \sqrt{\frac{2}{T}}\cos(\omega_0 t),\ 0 \le t \le T \rightarrow \begin{cases} s_1(t) = \sqrt{E}\,\psi_1(t) \\ s_2(t) = -\sqrt{\frac{E}{4}}\,\psi_1(t) \end{cases}$$





$h(t) = \psi_1(T - t)$

temel fonksiyonların enerjisi 1 olmak zorunda

$$a_i = \int_0^T s_i(t)\,\psi_1(t)\,dt \quad i = 1,2$$

$$a_1 = \int_0^T s_1(t)\cdot\psi_1(t)\,dt = \int_0^T \sqrt{E}\,\psi_1(t)\,\psi_1(t)\,dt = \sqrt{E}\underbrace{\int_0^T \psi_1^2(t)\,dt}_{E\psi_1} = \sqrt{E}$$

$$a_2 = \int_0^T s_2(t)\,\psi_1(t)\,dt = -\sqrt{\frac{E}{4}}\int_0^T \psi_1^2(t)\,dt = -\frac{\sqrt{E}}{2}$$

$$\gamma_0 = \frac{a_1 + a_2}{2} = \frac{\sqrt{E} - \sqrt{E}/2}{2} = \frac{\sqrt{E}}{4}$$

$P_B = ?$

$$P_h = Q\left(\frac{a_1 - a_2}{2\sigma_0}\right)$$

$$\sigma_0^2 = \frac{N_0}{2}\,E_n \quad (E_n = E\psi_1) \quad \text{ve } E_n = 1$$

$$\sigma_0^2 = \frac{N_0}{2}$$

$$P_B = Q\left(\frac{\sqrt{E} + \sqrt{E}/2}{2\sqrt{N_0/2}}\right) = Q\left(\sqrt{\frac{9}{5}\frac{E}{N_0}}\right)$$

Eb cinsinden ifade edelim

$$E_b = E_{s_1}P(s_1) + E_{s_2}P(s_2)$$

$$= E\,\frac{1}{2} + \frac{E}{4}\cdot\frac{1}{2} = \frac{5E}{8} \Rightarrow E = \frac{8E_b}{5}$$

$$\rightarrow P_B = Q\left(\sqrt{\frac{8}{5}\frac{E_b}{N_0}}\right)$$

II. yol

$$P_B = Q\left(\sqrt{\frac{E_b}{N_0}(1-\rho)}\right) \xrightarrow[\text{siz hesaplayın}]{\rho = -4/5} P_B = Q\left(\sqrt{\frac{9}{5}\frac{E_b}{N_0}}\right)$$

III. yol

$$P_B = Q\left(\sqrt{\frac{E_d}{N_0}}\right)$$

$$E_d = \int_0^T [s_1(t) - s_2(t)]^2\,dt = \frac{9}{4}E$$

$$\Rightarrow P_B = Q\left(\sqrt{\frac{9}{8}\frac{E}{N_0}}\right)$$